				[1]		
5	**(a)**		$B = 31$	**B1**		
				[1]		

Question			Answer	Marks	Guidance	
5	(b)		$71 = A\ln 12 + 31$	M1		
			$A = 16$	A1	CAO May see 16.097…A0	
				[2]		
5	(c)		79 or 78.9 or better	B1	from $16\ln 20 + 31$ and 78.9317 Condone 79.2… from unrounded value of A	
				[1]		
5	(d)		$120 = 16\ln X + 31$	M1	FT their A and B; allow sign error	
			exp(their $\frac{120-31}{16}$)	M1	Allow slip in rearrangement	
			awrt $X = 260$	A1	CAO $A = 16.097\ldots$ leads to 250, A0.	
				[3]		